

**RESEARCH ARTICLE**

# Doses of AQUAcare<sup>®</sup> probiotics on the duration of water quality conservation and effects on some zootechnical performances of *Clarias gariepinus* in above-ground breeding

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**Abstract**

This study aimed to evaluate the performance of probiotics on the shelf-life and the physicochemical quality of water and their effects on the growth of *Clarias gariepinus* juveniles. For this, five different treatments were used depending on the doses including T<sub>0</sub> (0g/m<sup>3</sup>), T<sub>1</sub> (1g/m<sup>3</sup>), T<sub>2</sub> (3g/m<sup>3</sup>), T<sub>3</sub> (5g/m<sup>3</sup>) and T<sub>4</sub> (7g/m<sup>3</sup>) in a completely randomized design repeated twice. The juveniles were raised separately for 90 days in tanks of 60 L each and installed under a building with a stocking density of 40 individuals. The fish were fed twice a day (8 a.m. and 7 p.m.) with LE GOUESSENT feed of 1.2 mm diameter at a rate of 7% of the biomass of their body weight during the first two weeks. This ration was readjusted every two weeks according to the increase in weight gain of the fish, as well as the feed and the size of the pellets. After 90 days of rearing, the obtained physicochemical parameters such as temperature (26.87 °C), pH (8.11), ammonia (0.081 mg/l) and nitrite (0.028 mg/l) remained stable. The significantly (p < 0.05) higher survival rate (65%) was recorded in T<sub>2</sub> (3g/m<sup>3</sup>) (p < 0.05). The significantly (p < 0.05) higher mean weight (111.16 ± 1.41 g) was obtained in individuals from the T<sub>3</sub> treatment (5 g/m<sup>3</sup>). The lowest food conversion index (0.55) (p < 0.05) was recorded in T<sub>1</sub> (1g/m<sup>3</sup>), as well as the best production cost (1669.29 FCFA/Kg). Concerning the weight-length relationship in all treatments, the results a proportional and increasing evolution

. Furthermore, the shelf life for probiotic treatments is on average 4 days with the highest value (4.93 days) obtained in T<sub>3</sub> (5g/m<sup>3</sup>) and the best percentage of water saved (85.82%) in T<sub>1</sub>(1g/m<sup>3</sup>).

**Practical application**

On the basis of the results obtained, for a clear improvement in the production of juvenile *Clarias gariepinus*, we recommend at a practical level to producers to use probiotics at the proportion of 1g/m<sup>3</sup> of probiotics for a better improvement in growth performance and a better percentage of water saving.

**Keywords:** Probiotics, water, *Clarias gariepinus*, zootechnical parameters.

**1. Introduction**

Over the past decades, the aquaculture industry has contributed significantly to the production of animal protein in almost every country in the world, accounting for approximately half of the seafood consumed each year (FAO, 2021).

Total fisheries and aquaculture production reached an unprecedented level of 223 million tons in 2022, an increase of 4.4 percent more than the previous record in 2020 (214 million tons) (FAO, 2024). This limited growth is mainly

due to a 4.4 percent decline in capture fisheries. This decline has been offset by continued growth in aquaculture, albeit at a slower pace over the past two years (FAO, 2022). In 2022 for the first time in history, aquaculture has become the main producer of aquatic animals (51% or 130.9 million tons) ahead of capture fishing (FAO, 2024). Based on the assumptions made, total fisheries and aquaculture production (excluding algae) is expected to continue to grow to reach 202 million tons in 2030. This represents an increase of 14 percent compared to 2020 and an additional volume of 24 million tons in absolute value (FAO, 2022). However, if the total volume continues to increase, projections indicate a slowdown in growth, in percentage and absolute value, compared to the figures recorded during the period 2010-2020 (23 percent and 33 million tons). Most of the increase in total global fisheries and aquaculture production will come from the aquaculture sector, whose production is expected to cross the 100 million ton mark for the first time in 2027 and then reach 106 million tons in 2030, be a total growth of 22 percent and almost 19 million tons compared to 2020 (FAO, 2022). Furthermore, access to water has become a powerful global economic issue which could become one of the main causes of international tensions. Local conflicts sometimes leading to wars are triggered over limited water resources. With the explosion of the world population and the increase in needs, these tensions could multiply in the future (Solar Impulse Foundation, 2019). Indeed, with demographic pressure, climate change, food production and energy needs, water is today a rare and poorly distributed resource in the world.

In this context, probiotics can play a vital role in aquaculture productivity by improving non-specific disease prevention and providing

pollution-free water sources through mineralization (Panigrahi *et al.*, 2010).

Water is one of the most precious resources on the planet. More than a billion people do not have access to a safe drinking water source and around 3 billion people face water shortages at least one month a year (Solar Impulse Foundation, 2019). Seen from the sky, it may seem excessive to talk about water shortages on a global scale. 1,385 million km<sup>3</sup> of water in fact covers three quarters of the surface of the earth, oceans, seas, lakes, rivers and groundwater and underground water combined (Barah, 2005). But it is salt water that dominates this vast expanse of water. Thus, of the 1,385 million km<sup>3</sup> of blue gold with which the planet abounds, only 34 million (or 2.5% of this total) is fresh water; But this poor availability of water resources does not alone explain the challenges we find ourselves facing today. Various studies carried-out on the subject affirm on the contrary that the few renewable water resources available today remain sufficient to meet the needs of humanity provided that they are used wisely, balanced and well supervised (Barah, 2005). The availability of renewable fresh water is in fact all the more poorly managed as the states of the planet cannot always boast of being well watered. Thus, many States have less than 1,700 m<sup>3</sup>/inhabitant/year of renewable fresh water, and therefore find themselves in a state of shortage. But this aspect cannot obscure one of the other major reasons why our water resources are dwindling, namely the inconsistent use made of them (Barah, 2005).

Water quality is the most crucial element of an aquaculture system. It is essential to the good health of fish added to the quality of the harvested product. To be successful, aquaculture requires quality water, the right amount at the right temperature (Veolia, 2019). Although a certain

amount of work has been carried-out to assess the impact of climate change on fisheries, little has yet been done in this area on aquaculture (Guichard *et al.*, 2017). Water exchange and biofiltration are traditional approaches to control hazardous metabolites in aquaculture (Jahangiri & Esteban, 2018).

Several systems and methods to improve water quality and treat aquaculture wastewater have also been proposed and implemented, including closed-loop aquaculture systems, biofloc technology, and aquaponics (Emerenciano *et al.*, 2017; Maucieri *et al.*, 2018). The most common approach to maintain water quality in aquaculture is frequent water exchange, which is expensive, time-consuming, and can introduce pathogens into culture systems (Devaraja *et al.*, 2013). Likewise, water scarcity and poor water quality are becoming an international problem, particularly in arid and semi-arid regions; therefore, the use of poor quality irrigation water is gaining importance in many countries in the global agricultural sector. Much research is underway to study the effectiveness of probiotics in water quality management in aquaculture (Hura *et al.*, 2018). Probiotics are often proposed as environmentally friendly alternatives to antibiotics (Jahangiri & Esteban, 2018).

In aquaculture, fish and microorganisms share an ecosystem. Therefore, fish welfare depends on the quality of the culture water. De B *et al.* (2014) described the beneficial effect of probiotics in pond water as an aid in the decomposition of organic matter, reduction of harmful organic waste that accumulates at the bottom of the pond, and reduction of phosphate and nitrogen compound levels when added to fish culture water. Nitrogenous waste is reduced through remineralization by probiotic populations (De B *et al.*, 2014; FAO, 2016b). Probiotics have been

used to improve the culture environment of shrimp in ponds by enhancing microbial and phytoplankton populations, resulting in a healthy ecosystem (FAO, 2016b). In particular, bacterial treatments aimed at improving water quality and production yield of *Penaeus monodon* resulted in a reduction of ammonia and nitrite in the rearing water (Gatesoupe, 1999). Higher levels of Gram-positive bacteria were used to reduce particulate organic carbon accumulation and minimize dissolved and particulate organic carbon accumulation during the shrimp culture cycle, while promoting more stable phytoplankton blooms through increased CO<sub>2</sub> production (Verschuere *et al.*, 2000). Furthermore, the introduction of *Bacillus* spp. near pond aerators enabled the degradation of organic matter, reduced chemical oxygen demand, thereby improving water quality and increasing shrimp harvests (FAO, 2016b).

The effects of probiotics on water quality in northern white shrimp (*Penaeus vannamei*) ponds were studied, and the results showed that commercial probiotics could reduce nitrogen and phosphorus concentrations in pond water (Zhou *et al.*, 2009). They also reported that there was no effect on water quality when *B. coagulans* SC8168 was used in shrimp (*P. vannamei*) ponds. It was explained that *B. coagulans* provided stable water quality parameters throughout the study period. The use of probiotics has been reported to directly regulate water quality, such as stable pH between 7.5 and 8.0 or healthy growth of shrimp (*P. monodon* and *Litopenaeus vannamei*) in Thailand (FAO, 2016b). El-Haroun *et al.* (2006) reported that a commercial probiotic based on *S. cerevisiae*, *Bacillus* spp., *Nitrosomonas* spp. and *Nitrobacter* spp. had the ability to reduce phosphate concentrations from 0.1105 to 0.0364 mg L<sup>-1</sup> and inorganic nitrogen from 3.74 to 1.79 mg L<sup>-1</sup>. In Nile tilapia, commercial probiotics

based on *Bacillus licheniformis* and *B. subtilis* were able to optimize dissolved oxygen and ammonia levels.

Seen from this angle probiotics would influence the physicochemical quality of water, such as the concentration of nitrite, ammonia, pH and temperature. Likewise, fish raised in systems using probiotics would exhibit faster growth, better feed conversion and greater survival. The use of probiotics would be economically beneficial in soilless aquaculture systems. With this in mind, this study was carried out with the general objective of contributing to the improvement of the production of *Clarias gariepinus* in an intensive system.

## 2. Materials and Methods

### 2.1. Materials

#### 2.1.1. Study zone and duration.

Mfou municipality is located at the intersection of Mbalmayo and Nkolafamba villages. It includes 08 villages headed by a 2<sup>nd</sup> degree leader. It is limited to the North by the District of Yaoundé IV (Mfoundi Division) and the Soa District (Mefou and Afamba Division to the South by the locality of Mbalmayo and Nkolmetet (Nyong and So'o Division); to the East and South-East by the Awa-Nkol-Afamba subdivisions (Mefou and Afamba Divisions), of Dzeng (Nyong and So'o Division); to the west by that of Bikok (Mefou and Akono Division). Its geographical coordinates are: a Latitude of 3° 52'12"North; a Longitude of 11°38'28 East and an Altitude of 694 m (PCD-Mfou, 2019).

#### 2.1.2. Biological materials

A total of 1200 juveniles of *Clarias gariepinus* with an average weight of approximately  $3 \pm 0.3$ g from the reception were divided into 05 treatments arranged in duplicate in 10 basins (60 L) filled to

a quarter (20 L) and gradually readjusted according to sizes; oxygenation was carried out using aerators.

Probiotic used (AQUAcare) is a Skreting brand product from SOCAVE in Yaoundé (total cavot). Its composition was as follows: Total bacillus (*B. amyloliquefaciens*, *B. pumilus*, *B. subtilis*)  $21.5 \times 10$  cfu/g *Pediococcus acidilacticiz*  $1.0 \times 10$  cfu/g CaCO<sub>3</sub>, (bulking agent).

#### 2.1.3. Accommodation and equipment

Ten circular basins of 60 liters each, covered with an anti-predator net operating in an open circuit and supplied with water from the well directly by a 7500 L/min Total brand pump were necessary for this experiment. The entire experimental device was installed in a room of approximately 100 m<sup>2</sup> without lighting but with an air pump to ensure aeration in the experimental tanks.

The treatments were administered as follows:

- Treatment T0: no probiotics;
- Treatment T1: probiotics at 1g/m<sup>3</sup>;
- Treatment T2: probiotics at 3g/m<sup>3</sup> (manufacturer's prescription);
- Treatment T3: probiotics at 5g/m<sup>3</sup>;
- Treatment T4: probiotics at 7g/m<sup>3</sup>.

LE GOUSSANT brand feed was used. The grain size of this feed was between 1 and 3 mm, purchased from SOCAVE in Yaoundé (total cavot).

Five treatments were distributed in 10 basins according to a completely randomized experimental design and replicated twice (5\*2).

## 2.2. Methods

### 2.2.1. Conduct of the test

To conduct the experiment, five treatments were tested, namely T0, T1, T2, T3 and T4 with doses of probiotics respectively of 0, 1, 3, 5 and 7g/m<sup>3</sup>. Each treatment was carried out in duplicate. The

experimental probiotics are inoculated at a double dose at each water renewal and the maintenance dose is applied every day until the next renewal.

During the experiment, the fishes were fed manually at a frequency of twice a day: 8 a.m. and 8 p.m. The rationing rate was determined based on the ration sheet defined by the feed used (LE GOUESSANT).

Control fishing was done every two weeks in the morning. Before fishing, the fish were left fasting to avoid the effect of the weight of the undigested feed on the weight of the animals. During these different fishing operations, several data were collected including the average weight evaluated by taking the individual weight of 10% per batch using an SF-400 brand scale with a sensitivity of 1g as well as the total and standard lengths using a graduated ruler. At the end of each control fishery the new quantities of feed were defined according to the average weight of the fish. The basins were emptied according to the treatments when the values of the physicochemical characteristics (water nitrites, ammonia) reached the critical value. These values were taken every evening using an Aquadene brand dosing kit as well; the temperature and pH were taken using an HTC-2 brand thermometer and an ATC pH meter.

### ***2.2.2. Characteristics and parameters studied***

During the experiment, the physicochemical and growth characteristics were studied. For the physicochemical characteristics it involved: temperature, concentrations of ammonia, nitrites and pH were measured daily.

The survival and growth characteristics studied were:

#### **Survival rate (SR)**

$$SR (\%) = 100 \times NF/Ni$$

NF = number of fish at the end of the experiment and Ni = number of fish at the start of the experiment (Cacot, 2006)..

#### **Live weight**

At the start of the test and every 14 days thereafter, fish from each experimental unit were weighed. The weekly weight gain was obtained by taking the difference between 2 consecutive average weekly live weights (Elegbe *et al.*, 2015).

**Average weight gain (AWG in g)** = final average fish weight (FAFW in g) - initial average fish weight (IAWG in g) (Elegbe *et al.*, 2015)

**Average daily gain (ADG in g/day)** = (FAWG-IAWG)/t With IAW = initial average weight (g), FAW = final average weight (g), t = duration of the experiment (in days) (Elegbe *et al.*, 2015).

**Specific growth rate (SGR in %day)** =  $[(\ln Pmf - \ln Pmi) / \text{rearing time (day)}] \times 100$ ; Pmi = initial average weight (g), Pmf = final average weight (g) Brown (1975).

**Feed conversion ratio (FCR)** = Quantity of feed distributed / Body mass gain (Elegbe *et al.*, 2015).

**Condition factor (K)** =  $W \times 100 / LT^3$  with W: weight (g), LT: Total length (cm) (Le Cren, 1951).

**Cost of feed intake** = cost of kg of feed x feed intake.

**Feed cost of production of one kg of fish (FPC in CFA francs)** = Purchase price of the feed  $\times$  CI (Elegbe *et al.*, 2015).

**Average duration of water conservation:** is the sum of the times of use of a treatment over the number of renewals of the latter.



## 2.3. Statistical analyzes

The data collected in a completely randomized device were processed in an Excel spreadsheet for physicochemical and economic characteristics. The zootechnical characteristics were subjected to the one-way analysis of variance (ANOVA) when there were differences between the means, the Duncan test was used to separate them at the 5% significance level. Statistical analyzes were carried out using the static analysis software SPSS (Statistical package of social science) version 25.0.

## 3. Results and discussion

### 3.1. Results

#### 3.1.1. Effect of probiotic dose (AQUAcare) on the physicochemical characteristics of water

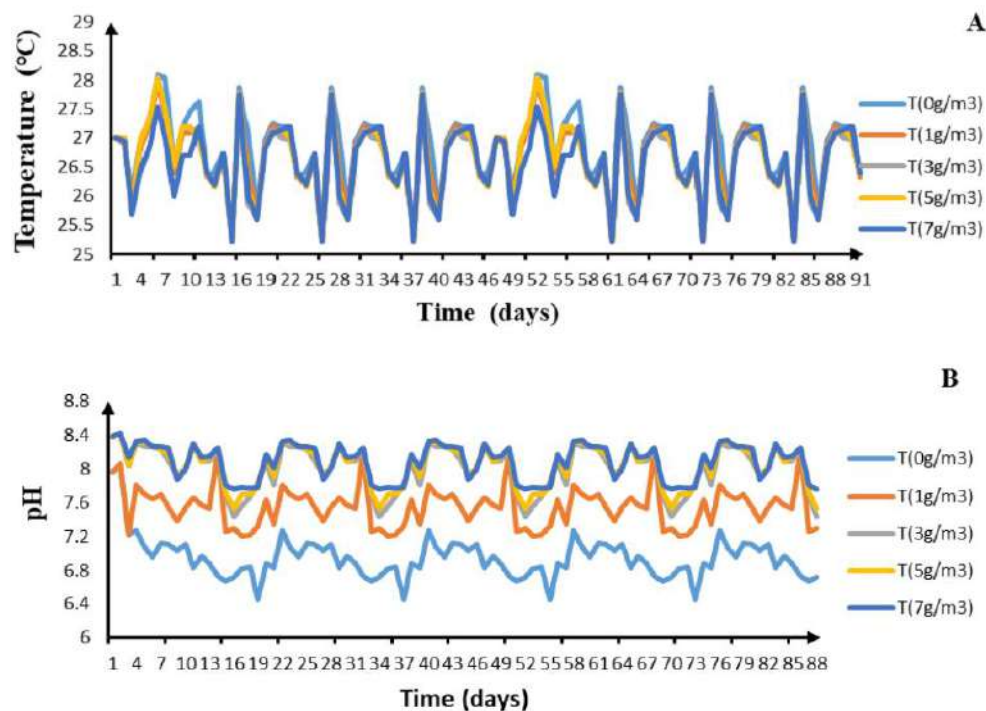
The evolution of the temperature and pH of the water are illustrated in Figure 1.

Likewise for pH, we observe a similar and distinct saw tooth evolution in all treatments. The pH varies between 7.26 and 8.23 for the treatments receiving probiotics and 6.73 and 7.38 for the control treatment without probiotics.

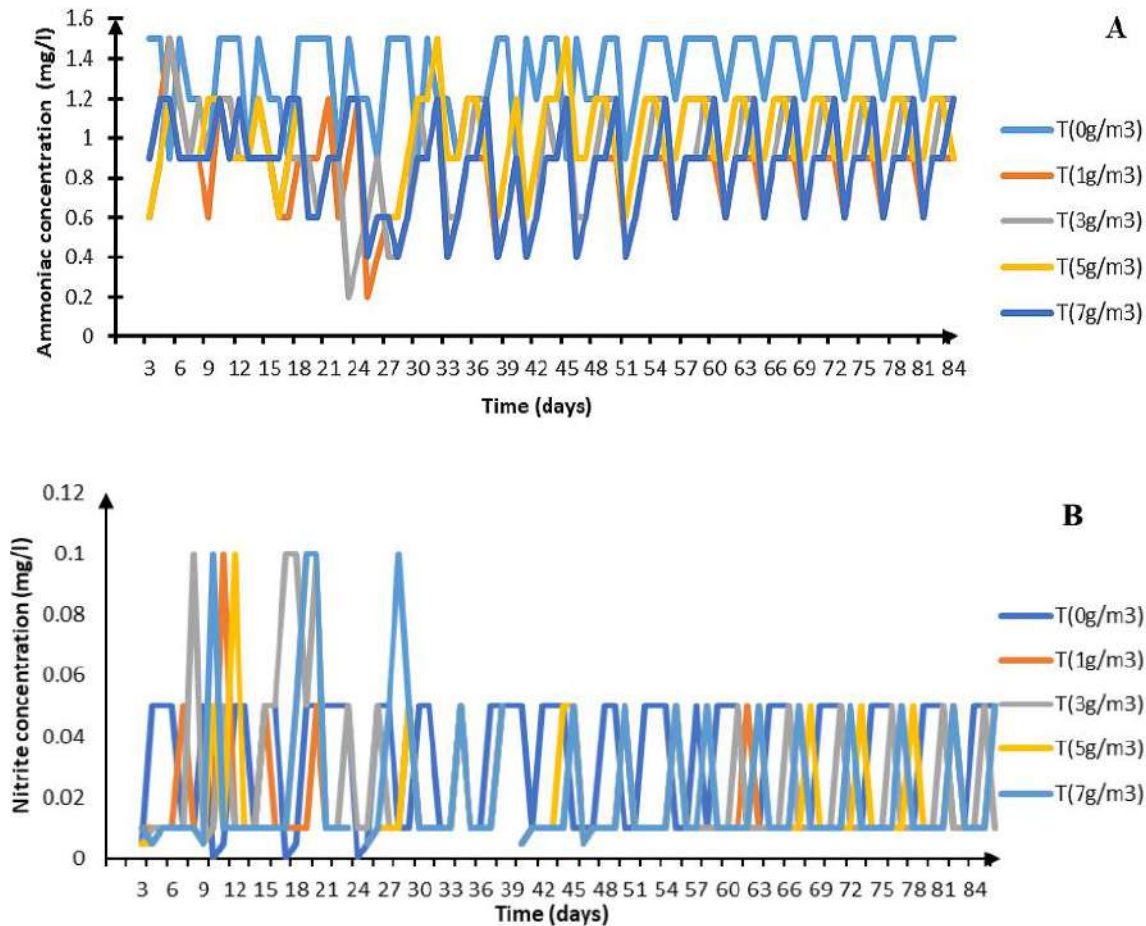
The temperature curves evolve in a similar and jagged manner in all the basins and the maximum value was recorded during the eighth week (27.34°C) in treatment T (0g/m<sup>3</sup>) and the minimum value recorded during the eighth dose (26.10 °C) in treatment T (3g/m<sup>3</sup>).

Likewise for pH, we observe a similar and distinct sawtooth evolution in all treatments. The pH varies between 7.26 and 8.23 for the treatments receiving probiotics and 6.73 and 7.38 for the control treatment without probiotics.

Figure 2 presents the evolution of ammonia and nitrite.



**Figure 1 (A-B):** Evolution of temperature (A) and pH (B) as a function of doses and time. °C: degree Celsius; T(0g/m<sup>3</sup>): Treatment with 0g/m<sup>3</sup> of probiotics; T (1g/m<sup>3</sup>): Treatment with 1g/m<sup>3</sup> of probiotics; T(3g/m<sup>3</sup>): Treatment with 3g/m<sup>3</sup> of probiotics; T(5g/m<sup>3</sup>): Treatment with 5g/m<sup>3</sup> of probiotics; T(7g/m<sup>3</sup>): Treatment with 7g/m<sup>3</sup> of probiotics.



**Figure 2 (A-B):** Evolution of ammonia (A) and nitrite (B) as a function of doses and time. mg: miligram ; T<sub>(0g/m<sup>3</sup>)</sub>: Treatment with 0g/m<sup>3</sup> of probiotics; T<sub>(1g/m<sup>3</sup>)</sub>: Treatment with 1g/m<sup>3</sup> of probiotics; T<sub>(3g/m<sup>3</sup>)</sub>: Treatment with 3g/m<sup>3</sup> of probiotics; T<sub>(5g/m<sup>3</sup>)</sub>: Treatment with 5g/m<sup>3</sup> of probiotics; T<sub>(7g/m<sup>3</sup>)</sub>: Treatment with 7g/m<sup>3</sup> of probiotics.

From Figure 2A we observe a similar sawtooth evolution for all treatments. However, the NH<sub>4</sub><sup>+</sup> concentration of the control treatment is above 1 mg/L, i.e. a value of 1.33 mg/L. Those of probiotic treatments remained below 1 mg/L variants between 0.81 mg/L for T(1g/m<sup>3</sup>) and 0.99 mg/L for T(5g/m<sup>3</sup>).

However, as shown in Figure 2B, we observe a similar and confused evolution of the curves of all the treatments. However, the NO<sub>2</sub> concentration of the control treatment averages 0.33 mg/L over the period, while those of the probiotic treatments vary between 0.018 mg/L for T(1g/m<sup>3</sup>) and 0.024 mg/L for T(3g/m<sup>3</sup>).

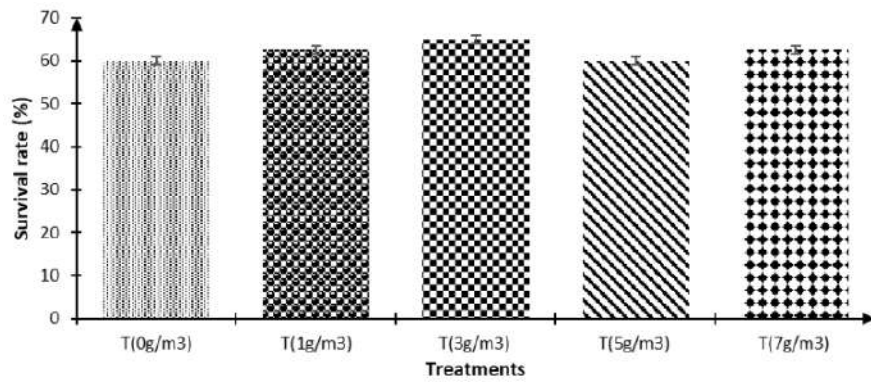
### 3.1.2 Effect of probiotic dose (AQUAcare) on zootechnical characteristics

#### 3.1.2.1 Survival Rate

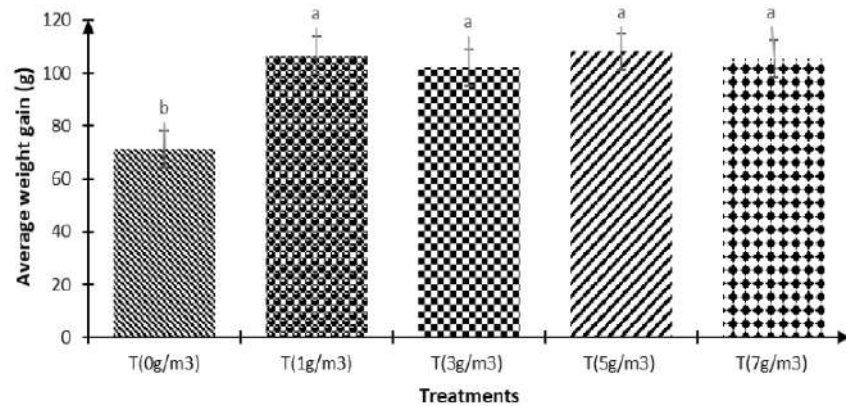
The variation in the Survival Rate is illustrated in Figure 3. The evolution of the survival rates during the experiment for each of the treatments is presented in Figure 3. These rates varied from 60% to 65% after 90 days of rearing. The treatment with the highest survival rate (65%) ( $p>0.05$ ) was recorded with T (3g/m<sup>3</sup>).

#### 3.1.2.2 Average Weight gain

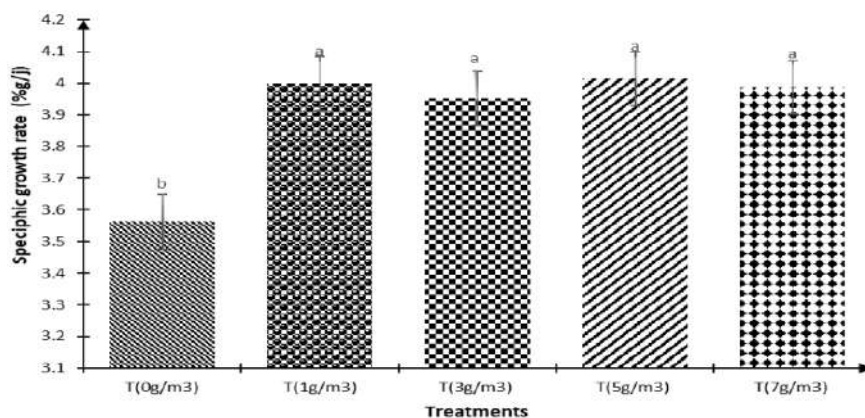
The variation in average weight is presented in Figure 4.



**Figure 3:** Variation in Survival Rates depending on the treatments. a, b: histograms bearing the same letters are significantly different ( $p > 0.05$ ).  $T_{(0g/m^3)}$ : Treatment with 0g/m<sup>3</sup> of probiotics;  $T_{(1g/m^3)}$ : Treatment with 1g/m<sup>3</sup> of probiotics;  $T_{(3g/m^3)}$ : Treatment with 3g/m<sup>3</sup> of probiotics;  $T_{(5g/m^3)}$ : Treatment with 5g/m<sup>3</sup> of probiotics;  $T_{(7g/m^3)}$ : Treatment with 7g/m<sup>3</sup> of probiotics

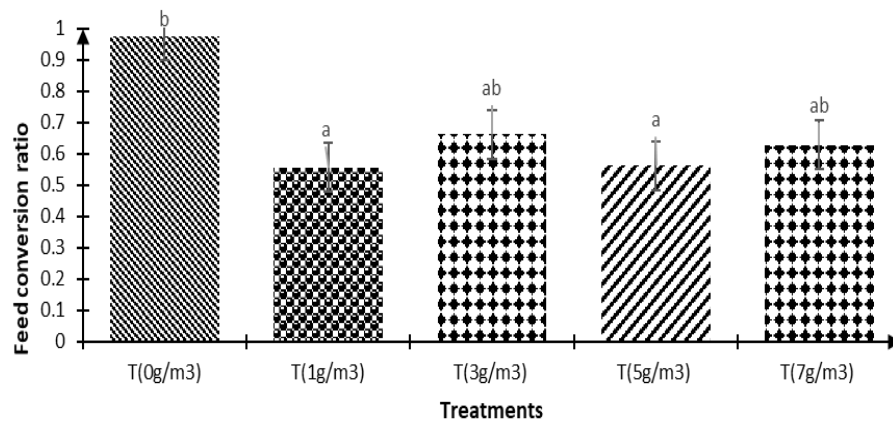


**Figure 4:** Variation in Weight Gain depending on the treatments.  $T_{(0g/m^3)}$ : Treatment with 0g/m<sup>3</sup> of probiotics;  $T_{(1g/m^3)}$ : Treatment with 1g/m<sup>3</sup> of probiotics;  $T_{(3g/m^3)}$ : Treatment with 3g/m<sup>3</sup> of probiotics;  $T_{(5g/m^3)}$ : Treatment with 5g/m<sup>3</sup> of probiotics;  $T_{(7g/m^3)}$ : Treatment with 7g/m<sup>3</sup> of probiotics



**Figure 5:** Specific Growth Rate depending on the treatments.  $T_{(0g/m^3)}$ : Treatment with 0g/m<sup>3</sup> of probiotics;  $T_{(1g/m^3)}$ : Treatment with 1g/m<sup>3</sup> of probiotics;  $T_{(3g/m^3)}$ : Treatment with 3g/m<sup>3</sup> of probiotics;  $T_{(5g/m^3)}$ : Treatment with 5g/m<sup>3</sup> of probiotics;  $T_{(7g/m^3)}$ : Treatment with 7g/m<sup>3</sup> of probiotics





**Figure 6:** Feed conversion ratio depending on treatments. T<sub>(0g/m<sup>3</sup>)</sub>: Treatment with 0g/m<sup>3</sup> of probiotics; T<sub>(1g/m<sup>3</sup>)</sub>: Treatment with 1g/m<sup>3</sup> of probiotics; T<sub>(3g/m<sup>3</sup>)</sub>: Treatment with 3g/m<sup>3</sup> of probiotics; T<sub>(5g/m<sup>3</sup>)</sub>: Treatment with 5g/m<sup>3</sup> of probiotics; T<sub>(7g/m<sup>3</sup>)</sub>: Treatment with 7g/m<sup>3</sup> of probiotics

According to Figure 4, the highest weight gains ( $P>0.05$ ) were recorded with the treatments receiving probiotics. However, treatment T (5g/m<sup>3</sup>) recorded the highest value ( $111.16\pm1.41$ g) non-significantly ( $P>0.05$ ) compared to the other treatments containing probiotics. Furthermore, the lowest average weight was recorded with the control treatment without probiotic T (0g/m<sup>3</sup>).

### 3.1.2.3. Specific Growth Rate

Figure 5 for its part presents the evolution of the Specific Growth Rates of the subjects for the different treatments. Specific Growth Rates followed the same trends as average weight gains, with the highest value recorded in T (5g/m<sup>3</sup>) ( $4.01\pm0.01$ ). Furthermore, the lowest Specific Growth Rate was recorded in the control treatment without probiotic T (0g/m<sup>3</sup>).

### 3.3.4. Feed Conversion ratio

Figure 6 for its part presents the feed conversion ratio of the different doses. The histograms in Figure 6 allow us to present the Feed Conversion Indexes whose values vary from 0.55 to 0.98 at Day 90 with the lowest value T (1g/m<sup>3</sup>) (0.55) significantly ( $P<0.05$ ) compared to the control

**Table 1:** Summary of zootechnical characteristics according to treatments

| Zootechnical characteristics |                                 |                                 |                                 |                                 |                                 |       |
|------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|-------|
| Characteristics              | T <sub>(0g/m<sup>3</sup>)</sub> | T <sub>(1g/m<sup>3</sup>)</sub> | T <sub>(3g/m<sup>3</sup>)</sub> | T <sub>(5g/m<sup>3</sup>)</sub> | T <sub>(7g/m<sup>3</sup>)</sub> | P     |
| Survival rate (%)            | 60.00±7.07                      | 62.50±3.53                      | 65.00±14.14                     | 60.00±14.14                     | 62.50±3.53                      | 0.981 |
| AW(g)                        | 74.22±7.73 <sup>a</sup>         | 109.75±1.95 <sup>b</sup>        | 105.09±1.90 <sup>b</sup>        | 111.16±1.41 <sup>b</sup>        | 108.50±0.70 <sup>b</sup>        | 0.001 |
| AWG (g)                      | 71.22±7.73 <sup>a</sup>         | 106.75±1.95 <sup>b</sup>        | 102.09±1.90 <sup>b</sup>        | 108.16±1.41 <sup>b</sup>        | 105.50±0.70 <sup>b</sup>        | 0.001 |
| ADWG (g)                     | 0.79±0.08 <sup>a</sup>          | 1.18±0.02 <sup>b</sup>          | 1.13±0.02 <sup>b</sup>          | 1.20±7.73 <sup>b</sup>          | 1.17±0.01 <sup>b</sup>          | 0.001 |
| SGR (%/g/j)                  | 3.56±0.11 <sup>a</sup>          | 3.99±0.01 <sup>b</sup>          | 3.95±0.02 <sup>b</sup>          | 4.01±0.01 <sup>b</sup>          | 3.98±0.01 <sup>b</sup>          | 0.002 |
| FCR                          | 0.98 <sup>b</sup>               | 0.55 <sup>a</sup>               | 0.71 <sup>ab</sup>              | 0.56 <sup>ab</sup>              | 0.63 <sup>a</sup>               | 0.041 |
| K                            | 0.62±0.02                       | 0.65±0.01                       | 0.78±0.20                       | 0.83±0.25                       | 0.83±0.22                       | 0.633 |

a, b: values bearing the same letters are significantly different (p>0.05). g: gram; SGR: specific growth rate; %g/d: percentage gram per day; %: percentage; AW: average weight; AWG: average weight gain; ADWG: average daily weight gain; FCR: Feed conversion ratio; K: condition factor. T<sub>(0g/m<sup>3</sup>)</sub>: Treatment with 0g/m<sup>3</sup> of probiotics; T<sub>(1g/m<sup>3</sup>)</sub>: Treatment with 1g/m<sup>3</sup> of probiotics; T<sub>(3g/m<sup>3</sup>)</sub>: Treatment with 3g/m<sup>3</sup> of probiotics; T<sub>(5g/m<sup>3</sup>)</sub>: Treatment with 5g/m<sup>3</sup> of probiotics; T<sub>(7g/m<sup>3</sup>)</sub>: Treatment with 7g/m<sup>3</sup> of probiotics.

a, b: values bearing the same letters are significantly different ( $p>0.05$ ). g: gram; SGR: specific growth rate; %/g/d: percentage gram per day; %: percentage; AW: average weight; AWG: average weight gain; ADWG: average daily weight gain; FCR: Feed conversion ratio; K: condition factor. T<sub>(0g/m<sup>3</sup>)</sub>: Treatment with 0g/m<sup>3</sup> of probiotics; T<sub>(1g/m<sup>3</sup>)</sub>: Treatment with 1g/m<sup>3</sup> of probiotics; T<sub>(3g/m<sup>3</sup>)</sub>: Treatment with 3g/m<sup>3</sup> of probiotics; T<sub>(5g/m<sup>3</sup>)</sub>: Treatment with 5g/m<sup>3</sup> of probiotics; T<sub>(7g/m<sup>3</sup>)</sub>: Treatment with 7g/m<sup>3</sup> of probiotics.

**Table 2:** Economic characteristics depending on doses

| Characteristics                                                                                                                                                                                                                                                                                                                                                                       | Economics parameters       |                            |                           |                           |                           |       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|----------------------------|---------------------------|---------------------------|---------------------------|-------|
|                                                                                                                                                                                                                                                                                                                                                                                       | T (0g/m <sup>3</sup> )     | T (1g/m <sup>3</sup> )     | T (3g/m <sup>3</sup> )    | T (5g/m <sup>3</sup> )    | T (7g/m <sup>3</sup> )    | P     |
| Total live weight (kg)                                                                                                                                                                                                                                                                                                                                                                | 1.77±0.12 <sup>b</sup>     | 2.74±0.07 <sup>a</sup>     | 2.72±0.04 <sup>a</sup>    | 2.67±0.06 <sup>a</sup>    | 2.71±0.03 <sup>a</sup>    | 0.001 |
| Feed quantity                                                                                                                                                                                                                                                                                                                                                                         | 2226.36±0.08 <sup>a</sup>  | 1830.84±0.12 <sup>b</sup>  | 1894.66±0.18 <sup>b</sup> | 1997.01±0.23 <sup>b</sup> | 1989.23±0.12 <sup>b</sup> | 0.001 |
| Total water (%)                                                                                                                                                                                                                                                                                                                                                                       | 100.00±0.00 <sup>a</sup>   | 14.18±1.00 <sup>b</sup>    | 15.34±4.00 <sup>b</sup>   | 14.47±3.00 <sup>b</sup>   | 14.76±3.00 <sup>b</sup>   | 0.001 |
| Production cost/kg                                                                                                                                                                                                                                                                                                                                                                    | 1751.08±30.00 <sup>c</sup> | 1669.29±23.00 <sup>d</sup> | 1987.01±80 <sup>b</sup>   | 2354.89±2.00 <sup>a</sup> | 2523.20±1.00 <sup>a</sup> | 0.001 |
| Number of days of storage                                                                                                                                                                                                                                                                                                                                                             | 1.00±00 <sup>b</sup>       | 4.64±0.10 <sup>a</sup>     | 4.23±0.50 <sup>a</sup>    | 4.93±0.70 <sup>a</sup>    | 4.86±0.60 <sup>a</sup>    | 0.001 |
| Kg : kilogram T (0g/m <sup>3</sup> ): Treatment with 0g/m <sup>3</sup> of probiotics; T (1g/m <sup>3</sup> ): Treatment with 1g/m <sup>3</sup> of probiotics; T (3g/m <sup>3</sup> ): Treatment with 3g/m <sup>3</sup> of probiotics; T (5g/m <sup>3</sup> ): Treatment with 5g/m <sup>3</sup> of probiotics; T (7g/m <sup>3</sup> ): Treatment with 7g/m <sup>3</sup> of probiotics. |                            |                            |                           |                           |                           |       |

treatment (0.98). However, the consumption index was comparable ( $P>0.05$ ) between the treatments receiving the probiotics. Except for survival rate and condition factor (K), all studied characteristics were significantly ( $P>0.05$ ) affected by the use of probiotics in rearing water. However, the highest mean weight, mean and weekly weight gain and specific growth rate were

recorded with the treatment containing 5 grams of probiotic per cubic meter of water (T (5g/m<sup>3</sup>)).

Table 2 presents the economic characteristics according to the different doses of probiotic. The total live weight during the experiment varies from 1.77 for T(0g/m<sup>3</sup>) to 2.74 for T(1g/m<sup>3</sup>) against feed consumption ranging from 1830.84 in T(1g/m<sup>3</sup>) to 2226.36 in T(0g/m<sup>3</sup>). The combination of the different costs involved in the experiment (cost of energy due to aeration and the pump; cost of the probiotic and feed) records the best production cost in T(1g/m<sup>3</sup>) (1669 Fcfa) and the highest in T(5g/m<sup>3</sup>) (2355 Fcfa) with regard to probiotic treatments compared to 1751 F in T(0g/m<sup>3</sup>). As for the maximum conservation period of water, it is on average 4 days for probiotic treatments against a daily renewal in T (0g/m<sup>3</sup>) which makes it possible to reduce the volume of water used to 14, 18% in T(1g/m<sup>3</sup>), 14.47% in T(5g/m<sup>3</sup>), 14.76% in T(7g/m<sup>3</sup>), 15.34% in T(3g/m<sup>3</sup>) compared to T(0g/m<sup>3</sup>) (100%) which are presented in table 2.

### 3.2. Discussion

The average temperature values are relatively identical in the experimental treatments. This could be due to the microclimate of the farm. They are in the recommended optimum range between 26 and 30 °C (FAO, 2011) in *Clarias gariepinus*, as confirmed by [Beggui & Zaidi, \(2019\)](#) whose values vary from 27.5 to 29.2.

The pH values obtained remained within the ranges favorable to the growth of freshwater fish. Indeed, no extreme, lethal or limiting value, basic or acidic, was observed during the experiment and is within the optimal range (6.5 and 8.5) recommended according to [FAO \(2011\)](#) for a good growth in catfish as obtained by [Sow \(2021\)](#) (7 to 8.5).

Generally, Ammonia values were much lower after flushing in the different experimental treatments and gradually increased during the water conservation period to stabilize at 1.20mg/L before gradually falling back to 0.9. Unlike the witness, this value increases with the length of the period and presents a constantly increasing curve. This variation would be related to the action of probiotics. The average values of probiotic treatments remained below 1 mg/L like the aquaponic systems of [Beggui & zaidi \(2019\)](#) and bioflocs of [Sow \(2021\)](#). Likewise the value of the control T(0g/m<sup>3</sup>) (1, 34 mg/L) that of [Beggui & zaidi \(2019\)](#) i.e. 2 mg/l rises above 1 mg/L. Discuss the changes observed by explaining them.

Nitrite values were generally much lower in experimental treatments after water drain and increased gradually and consistently during the water conservation period to stabilize at 0.01 mg/L depending on the duration of the period before to continued its growth. The average values were relatively close to the nitrite content required for good growth which should not exceed 0.5 mg/L ([FAO, 2011](#)) as obtained by [Sow \(2021\)](#). Indeed these values (between 0.018 mg/L and 0.33 mg/L) are close to that obtained by [Opiyo & Adhiambo \(2020\)](#) (0.03 mg/L). However, [Beggui & zaidi, \(2019\)](#) reported a nitrite value greater than 1 mg/L. Discuss the changes observed by explaining them.

The results of this experiment did not reveal differences in survival rates between treatments (> 60%) which is in line with previous work by [Compaore et al. \(2016\)](#) who obtained varying rates between 47.87% to 68.87 but these results were better than those found by [Toudjani et al. \(2024\)](#) which was 55%. The increase in survival rate may be due to the use of probiotics. The lack of significant differences reflects normal survival of the subjects. On the other hand, they are

lower than the results obtained by [Mohamed et al. \(2020\)](#) which was ranged between 77-94.75%. In this case the apparent difference could be the result of species divergence because the two scenarios probiotics are used. It should be noted that the greatest reductions in numbers observed were directly linked to cannibalism since the fish found dead showed traces of aggression. Proof that they were attacked by their conspecifics.

As for the average gains obtained during this study, they are better in probiotic treatments compared to T(0g/m<sup>3</sup>). This is in line with the previous work of [Mohamed et al. \(2020\)](#) which demonstrates the effectiveness of probiotics on the control and those despite the species divergence. Likewise, these results corroborate the work of [Compaore et al. \(2016\)](#); [Toudjani et al. \(2024\)](#) demonstrating that the subjects have normal growth

The evolution of the specific growth rate was similar to that of weight gain as reported by [Mohamed et al. \(2020\)](#) for the same species. Although the values was the same, the values were lower than those obtained by [Compaore et al. \(2016\)](#) (10.79 to 12.02) for *C. gariepinus*.

The feed conversion ratio (FCR) which is a criterion used in zootechnics to measure the efficiency of converting a feed into a given production unit of live weight shows that probiotic treatments (T(1g/m<sup>3</sup>), T (3g/m<sup>3</sup>), T(5g/m<sup>3</sup>), T(7g/m<sup>3</sup>)) enhanced the food better than the control treatment (without probiotic) with the best value in T(1g/m<sup>3</sup>) (0.55). The results obtained with probiotics were better than those of [Compaore et al. \(2016\)](#) and [Toudjani et al. \(2024\)](#) which were respectively ranged from 0.7 to 0.82 and 1.42 to 1.98. This difference could be the result of the use of probiotics which acts on the microbiota.

On the other hand, they are also better than those of Mohamed et al. (2020) who obtained varying rates between 1.49 and 2.11 with probiotics on tilapia, so the species could be the variation factor.

#### 4. Conclusion

The objective of this study was to evaluate the effect of the dose of AQUAcare\* probiotics on the duration of conservation of water quality and some zootechnical performances of *Clarias gariepinus* in soilless rearing. It appears that the level of use of probiotics which allowed to obtain the best zootechnical performances is 5g/m<sup>3</sup> of probiotics and the level of use of probiotics which allowed to obtain the highest physicochemical performances (P>0.05) is 1g/m<sup>3</sup> of probiotics.

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#### Conflict of interest

The authors declare that there are no conflicts of interest.

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